

CS2220: Introduction to Computational Biology

Greedy Motif Search

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Adapted with modifications from lecture notes prepared by Phillip Compeau,

Bioinformatics Algorithms: An Active Learning Approach

Outline

- Motif Finding Problem - revisited
- Greedy Profile Motif Search
- Gibbs Sampler

What are motifs?

Cells can have the same DNA but behave differently because the rates of **expression** of genes (rate at which genes are converted to protein) can differ.

But how can genes be turned off/on?

Transcription factor: master regulatory protein that turns other genes on or off by binding to a specific short DNA interval called a **regulatory motif** 600-1000 nt preceding the gene.

The Motif Finding Problem

Given a list of t sequences each of length n , find the “best” pattern of length l that appears in each of the t sequences.

A New Motif Finding Approach

- **Previously:** we solved the Motif Finding Problem using a Branch and Bound technique.
- **Now:** **randomly** select possible locations and find a way to **greedily** change those locations until we have converged to the hidden motif.

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Randomized Algorithms

- Randomized algorithms make random rather than deterministic decisions.
- The main advantage is that no input can reliably produce worst-case results because the algorithm runs differently each time.
- These algorithms are commonly used in situations where no exact and fast algorithm is known.

Profiles Revisited

- Let $\mathbf{s}=(s_1,\dots,s_t)$ be the set of starting positions for l -mers in our t sequences.
- The substrings corresponding to these starting positions will form:
 - $t \times l$ *alignment matrix* and
 - $4 \times l$ *profile matrix* $\mathbf{P}$.

From Motifs to a Profile Matrix

Profile Matrix: formed by taking the frequency of symbols in each column of *Motifs*.

Motifs	T	C	G	G	G	G	g	T	T	T	t	t	
	c	C	G	G	t	G	A	c	T	T	a	C	
	a	C	G	G	G	G	A	T	T	T	t	C	
	T	t	G	G	G	G	A	c	T	T	t	t	
	a	a	G	G	G	G	A	c	T	T	C	C	
	T	t	G	G	G	G	A	c	T	T	C	C	
	T	C	G	G	G	G	A	T	T	c	a	t	
	T	C	G	G	G	G	A	T	T	c	C	t	
	T	a	G	G	G	G	A	a	c	T	a	C	
	T	C	G	G	G	t	A	T	a	a	C	C	
PROFILE(Motifs)	A:	.2	.2	0	0	0	0	.9	.1	.1	.1	.3	0
	C:	.1	.6	0	0	0	0	0	.4	.1	.2	.4	.6
	G:	0	0	1	1	.9	.9	.1	0	0	0	0	0
	T:	.7	.2	0	0	.1	.1	0	.5	.8	.7	.3	.4

Scoring Strings with a Profile

- $Prob(\mathbf{a}|\mathbf{P})$ is defined as the probability that an l -mer $\mathbf{a}$ was created by the Profile $\mathbf{P}$.

$$Prob(\mathbf{a}|\mathbf{P}) = \prod_{i=1}^l p_{a_i, i}$$

Scoring Strings with a Profile

(cont'd)

Given a profile: $\mathbf{P} =$

A	1/2	7/8	3/8	0	1/8	0
C	1/8	0	1/2	5/8	3/8	0
T	1/8	1/8	0	0	1/4	7/8
G	1/4	0	1/8	3/8	1/4	1/8

The probability of the consensus string:

$Prob(aaacct|\mathbf{P}) = ???$

Scoring Strings with a Profile

(cont'd)

Given a profile: $\mathbf{P} =$

A	1/2	7/8	3/8	0	1/8	0
C	1/8	0	1/2	5/8	3/8	0
T	1/8	1/8	0	0	1/4	7/8
G	1/4	0	1/8	3/8	1/4	1/8

The probability of the consensus string:

$$Prob(aaacct|\mathbf{P}) = 1/2 \times 7/8 \times 3/8 \times 5/8 \times 3/8 \times 7/8 = .033646$$

Scoring Strings with a Profile

(cont'd)

Given a profile: $P =$

A	1/2	7/8	3/8	0	1/8	0
C	1/8	0	1/2	5/8	3/8	0
T	1/8	1/8	0	0	1/4	7/8
G	1/4	0	1/8	3/8	1/4	1/8

Probability of a different string:

$$Prob(atacag|P) = 1/2 \times 1/8 \times 3/8 \times 5/8 \times 1/8 \times 1/8 = .001602$$

From a Profile Matrix to Motifs

The **probability** of a k -mer *text* for a given profile matrix *Profile*, written $\Pr(\text{text} | \text{Profile})$, is the product of profile matrix values for each symbol of *text*.

Answer: It increases, so we should be looking for k -mers that have large probabilities $\Pr(\text{text} | \text{Profile})$.

$$\Pr(\text{ACGGGGATTACC} | \text{Profile}) = .2 \cdot .6 \cdot 1 \cdot 1 \cdot .9 \cdot .9 \cdot .9 \cdot .5 \cdot .8 \cdot .1 \cdot .4 \cdot .6 \\ = 0.000839808$$

PROFILE(<i>Motifs</i>)	A:	.2	.2	0	0	0	0	.9	.1	.1	.1	.3	0
	C:	.1	.6	0	0	0	0	0	.4	.1	.2	.4	.6
	G:	0	0	1	1	.9	.9	.1	0	0	0	0	0
	T:	.7	.2	0	0	.1	.1	0	.5	.8	.7	.3	.4

P-Most Probable l -mer

- **P**-most probable l -mer for a sequence:
 an l -mer in that sequence which has the highest probability of being created from the profile **P**.

P =

A	1/2	7/8	3/8	0	1/8	0
C	1/8	0	1/2	5/8	3/8	0
T	1/8	1/8	0	0	1/4	7/8
G	1/4	0	1/8	3/8	1/4	1/8

Given a sequence = ctataaaccttacatc, find the P-most probable l -mer

P-Most Probable l -mer (cont'd)

A	1/2	7/8	3/8	0	1/8	0
C	1/8	0	1/2	5/8	3/8	0
T	1/8	1/8	0	0	1/4	7/8
G	1/4	0	1/8	3/8	1/4	1/8

Find the $Prob(a|P)$ of every possible 6-mer:

First try: c t a t a a a c c t t a c a t c

Second try: c t a t a a a c c t t a c a t c

Third try: c t a t a a a c c t t a c a t c

-Continue this process to evaluate every possible 6-mer

P-Most Probable l -mer (cont'd)

Compute $prob(a|P)$ for every possible 6-mer:

String, Highlighted in Red	Calculations	$prob(a P)$
ctataa ac cttacat	$1/8 \times 1/8 \times 3/8 \times 0 \times 1/8 \times 0$	0
ctataa aa ccttacat	$1/2 \times 7/8 \times 0 \times 0 \times 1/8 \times 0$	0
ctataa ac cttacat	$1/2 \times 1/8 \times 3/8 \times 0 \times 1/8 \times 0$	0
ctataa ac cttacat	$1/8 \times 7/8 \times 3/8 \times 0 \times 3/8 \times 0$	0
ctataa ac cttacat	$1/2 \times 7/8 \times 3/8 \times 5/8 \times 3/8 \times 7/8$	.0336
ctataa ac cttacat	$1/2 \times 7/8 \times 1/2 \times 5/8 \times 1/4 \times 7/8$	.0299
ctataa ac cttacat	$1/2 \times 0 \times 1/2 \times 0 \times 1/4 \times 0$	0
ctataa aa ccttacat	$1/8 \times 0 \times 0 \times 0 \times 0 \times 1/8 \times 0$	0
ctataa ac cttacat	$1/8 \times 1/8 \times 0 \times 0 \times 3/8 \times 0$	0
ctataa ac cttacat	$1/8 \times 1/8 \times 3/8 \times 5/8 \times 1/8 \times 7/8$	.0004

P-Most Probable l -mer (cont'd)

P-Most Probable 6-mer in the sequence is aaacct:

String, Highlighted in Red	Calculations	$Prob(a P)$
ctataa ac cttacat	$1/8 \times 1/8 \times 3/8 \times 0 \times 1/8 \times 0$	0
ctataaa c cttacat	$1/2 \times 7/8 \times 0 \times 0 \times 1/8 \times 0$	0
ctataa ac cttacat	$1/2 \times 1/8 \times 3/8 \times 0 \times 1/8 \times 0$	0
ctataa ac cttacat	$1/8 \times 7/8 \times 3/8 \times 0 \times 3/8 \times 0$	0
ctataaaccttacat	$1/2 \times 7/8 \times 3/8 \times 5/8 \times 3/8 \times 7/8$	.0336
ctata aac cttacat	$1/2 \times 7/8 \times 1/2 \times 5/8 \times 1/4 \times 7/8$	.0299
ctataa ac cttacat	$1/2 \times 0 \times 1/2 \times 0 \times 1/4 \times 0$	0
ctataaa c cttacat	$1/8 \times 0 \times 0 \times 0 \times 0 \times 1/8 \times 0$	0
ctataa ac cttacat	$1/8 \times 1/8 \times 0 \times 0 \times 3/8 \times 0$	0
ctataa ac cttacat	$1/8 \times 1/8 \times 3/8 \times 5/8 \times 1/8 \times 7/8$	.0004

P-Most Probable l -mer (cont'd)

aaacct is the P-most probable 6-mer in:

ctat**aaacct**tacatc

because $Prob(aaacct|P) = .0336$ is greater than the $Prob(a|P)$ of any other 6-mer in the sequence.

P-Most Probable l -mers in Many Sequences

- Find the **P**-most probable l -mer in **each** of the sequences.

P=

A	1/2	7/8	3/8	0	1/8	0
C	1/8	0	1/2	5/8	3/8	0
T	1/8	1/8	0	0	1/4	7/8
G	1/4	0	1/8	3/8	1/4	1/8

ctataaacgttacatc
 atagcgattcgactg
 cagcccagaaccct
 cggtataccttacatc
 tgcattcaatagctta
 tatcctttccactcac
 ctccaaatcctttaca
 ggatcatcctttatcct

P-Most Probable l -mers in Many Sequences (cont'd)

1	a	a	a	c	g	t
2	a	t	a	g	c	g
3	a	a	c	c	c	t
4	g	a	a	c	c	t
5	a	t	a	g	c	t
6	g	a	c	c	t	g
7	a	t	c	c	t	t
8	t	a	c	c	t	t
A	5/8	5/8	4/8	0	0	0
C	0	0	4/8	6/8	4/8	0
T	1/8	3/8	0	0	3/8	6/8
G	2/8	0	0	2/8	1/8	2/8

ctata**aaacg**ttacatc

atagcgattcgactg

cagcccaga**aaccct**

cggt**gaac**ttacatc

tgcattca**atag**ctta

tgtcctgtccactcac

ctccaa**atc**ctttaca

ggct**tac**ctttatcct

P-Most Probable l -mers form a new profile

Comparing New and Old Profiles

1	a	a	a	c	g	t
2	a	t	a	g	c	g
3	a	a	c	c	c	t
4	g	a	a	c	c	t
5	a	t	a	g	c	t
6	g	a	c	c	t	g
7	a	t	c	c	t	t
8	t	a	c	c	t	t
A	5/8	5/8	4/8	0	0	0
C	0	0	4/8	6/8	4/8	0
T	1/8	3/8	0	0	3/8	6/8
G	2/8	0	0	2/8	1/8	2/8

Profile from P-Most Probable *l*-mers

Red – frequency increased
Blue – frequency decreased

A	1/2	7/8	3/8	0	1/8	0
C	1/8	0	1/2	5/8	3/8	0
T	1/8	1/8	0	0	1/4	7/8
G	1/4	0	1/8	3/8	1/4	1/8

Profile from the sequences

We don't have profiles!

STOP and Think: How to create the profile at the first place?

From a Profile Matrix to Motifs

Given profile matrix of strings Dna , $Motifs(Profile)$ is the strings formed by taking the most probable k -mer in each string (after pseudocounts).

So we can move from one collection of motifs in the search space to the next by taking two steps:

$$Motifs \rightarrow Profile(Motifs) \rightarrow Motifs(Profile(Motifs))$$

RandomizedMotifSearch starts with an arbitrary $Motifs$ and continues taking these steps until $Score(Motifs)$ stops improving.

Greedy Profile Motif Search

- ❖ Use P -Most probable l -mers to adjust start positions until we reach a “best” profile; this is the motif.
- 1) Select random starting positions.
- 2) Create a profile $\mathbf{P}$ from the substrings at these starting positions.
- 3) Find the $\mathbf{P}$ -most probable l -mer $\mathbf{a}$ in each sequence and change the starting position to the starting position of $\mathbf{a}$.
- 4) Compute a new profile based on the new starting positions after each iteration and proceed until we cannot increase the score anymore.

GreedyProfileMotifSearch Algorithm

1. GreedyProfileMotifSearch(DNA, t, n, l)
2. Randomly select starting positions $s=(s_1, \dots, s_t)$ from DNA
3. $bestScore \leftarrow 0$
4. **while** $Score(s, DNA) > bestScore$
5. Form profile P from s
6. $bestScore \leftarrow Score(s, DNA)$
7. **for** $i \leftarrow 1$ **to** t
8. Find a P -most probable l -mer a from the i^{th} sequence
9. $s_i \leftarrow$ starting position of a
10. **return** $bestScore$

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How Can a Randomized Algorithm Perform Well?

If the strings in *DNA* were truly random, then we would expect a uniform profile matrix, which is useless for motif finding...

A:	0.25	0.25	0.25	0.25
C:	0.25	0.25	0.25	0.25
G:	0.25	0.25	0.25	0.25
T:	0.25	0.25	0.25	0.25

How Can a Randomized Algorithm Perform Well?

If we were very lucky, then we might get a profile matrix that is much less uniform. (Say that the true motif is “ACGT”.)

A:	0.8	0.0	0.0	0.2
C:	0.0	0.6	0.2	0.0
G:	0.2	0.2	0.8	0.0
T:	0.0	0.2	0.0	0.8

- There is little chance that our guess will be close to an optimal motif, meaning it will take a very long time to find the optimal motif.

How Can a Randomized Algorithm Perform Well?

In practice, we are hoping that some of our randomized initial motifs find some signal, and start to point us toward the correct motifs.

A:	0.4	0.2	0.2	0.2
C:	0.2	0.4	0.2	0.2
G:	0.2	0.2	0.4	0.2
T:	0.2	0.2	0.2	0.4

By taking the *Profile*-most probable k -mer in each string, we have a greater chance of moving toward “ACGT” (although this is not certain).

How Can a Randomized Algorithm Perform Well?

Exercise: What are the chances that ten randomly selected 15-mers from ten 600-nucleotide long strings capture at least one implanted 15-mer?

Answer: The likelihood of missing all of them is $(1 - 1/586)^{10} \sim 0.983$, so with thousands of runs, we will almost certainly capture some collections of motifs with one or more “hits”.

Issues with Randomized Motif Search

By taking only the most probable k -mer at each step, **RandomizedMotifSearch** is very "rigid", as it can move only in one direction. (In fact, it's only randomization is in the initial choice of k -mers.)

Idea: Perhaps we could allow moving from one collection of motifs to another based on randomization.

Gibbs Sampling

- In practice, GreedyProfileMotifSearch algorithm is run many times with the hope that random starting positions will be close to the optimum solution simply by chance.
- However, we can improve the algorithm by introducing **Gibbs Sampling**, an iterative procedure that discards one l -mer after each iteration and replaces it with a new one.
- Gibbs Sampling proceeds more slowly and chooses new l -mers at random increasing the odds that it will converge to the correct solution.

How Gibbs Sampling Works

- 1) Randomly choose starting positions
 $\mathbf{s} = (s_1, \dots, s_t)$ and form the set of l -mers associated with these starting positions.
- 2) Randomly choose one of the t sequences.
- 3) Create a profile $\mathbf{P}$ from the other $t - 1$ sequences.
- 4) For each position in the removed sequence, calculate the probability that the l -mer starting at that position was generated by $\mathbf{P}$.
- 5) Choose a new starting position for the removed sequence at random based on the probabilities calculated in step 4.
- 6) Repeat steps 2-5 until there is no improvement

Gibbs Sampling: an Example

Input:

$t = 5$ sequences, motif length $l = 8$

1. GTAAACAATATTTATAGC
2. AAAATTTACCTCGCAAGG
3. CCGTACTGTCAAGCGTGG
4. TGAGTAAACGACGTCCCA
5. TACTTAACACCCTGTCAA

Gibbs Sampling: an Example

1) Randomly choose starting positions,
 $s=(s_1, s_2, s_3, s_4, s_5)$ in the 5 sequences:

$s_1=7$	GTAAACAATATTTATAGC
$s_2=11$	AAAATTTACCTTAGAAGG
$s_3=9$	CCGTACTGTCAAGCGTGG
$s_4=4$	TGAGTAAACGACGTCCCA
$s_5=1$	TACTTAACACCCTGTCAA

Gibbs Sampling: an Example

2) Choose one of the sequences at random:

Sequence 2: AAAATTTACCTTAGAAGG

$s_1=7$	GTAAACAATATTTATAGC
$s_2=11$	AAAATTTACCTTAGAAGG
$s_3=9$	CCGTACTGTCAAGCGTGG
$s_4=4$	TGAGTAAACGACGTCCCA
$s_5=1$	TACTTAACACCCTGTCAA

Gibbs Sampling: an Example

2) Choose one of the sequences at random:

Sequence 2: AAAATTTACCTTAGAAGG

$s_1=7$ GTAAACAATATTTATAGC

$s_3=9$ CCGTACTGTCAAGCGTGG

$s_4=4$ TGAGTAAACGACGTCCCA

$s_5=1$ TACTTAACACCCTGTCAA

Gibbs Sampling: an Example

3) Create profile P from l -mers in remaining 4 sequences:

1	A	A	T	A	T	T	T	A
3	T	C	A	A	G	C	G	T
4	G	T	A	A	A	C	G	A
5	T	A	C	T	T	A	A	C
A	1/4	2/4	2/4	3/4	1/4	1/4	1/4	2/4
C	0	1/4	1/4	0	0	2/4	0	1/4
T	2/4	1/4	1/4	1/4	2/4	1/4	1/4	1/4
G	1/4	0	0	0	1/4	0	3/4	0
Consensus String	T	A	A	A	T	C	G	A

Gibbs Sampling: an Example

- 4) Calculate the $prob(a|P)$ for every possible 8-mer in the removed sequence:

Strings Highlighted in Red	$prob(a P)$
AAAATTTACCTTAGAAGG	.000732
AAAATTTACCTTAGAAGG	.000122
AAAATTTACCTTAGAAGG	0
AAAATTTACCTTAGAAGG	0
AAAATTTACCTTAGAAGG	0
AAAATTTACCTTAGAAGG	0
AAAATTTACCTTAGAAGG	0
AAAATTTACCTTAGAAGG	.000183
AAAATTTACCTTAGAAGG	0
AAAATTTACCTTAGAAGG	0
AAAATTTACCTTAGAAGG	0

Gibbs Sampling: an Example

5) Create a distribution of probabilities of l -mers $prob(a|P)$, and randomly select a new starting position based on this distribution.

a) To create this distribution, divide each probability $prob(a|P)$ by the lowest probability:

Starting Position 1: $prob(\text{AAAATTTA} | P) = .000732 / .000122 = 6$

Starting Position 2: $prob(\text{AAATTTAC} | P) = .000122 / .000122 = 1$

Starting Position 8: $prob(\text{ACCTTAGA} | P) = .000183 / .000122 = 1.5$

Ratio = 6 : 1 : 1.5

Turning Ratios into Probabilities

b) Define probabilities of starting positions according to computed ratios

Probability (Selecting Starting Position 1): $6/(6+1+1.5)= 0.706$

Probability (Selecting Starting Position 2): $1/(6+1+1.5)= 0.118$

Probability (Selecting Starting Position 8): $1.5/(6+1+1.5)=0.176$

Gibbs Sampling: an Example

c) Select the start position according to computed ratios:

$P(\text{selecting starting position 1}):$.706

$P(\text{selecting starting position 2}):$.118

$P(\text{selecting starting position 8}):$.176

Gibbs Sampling: an Example

Assume we select the substring with the highest probability – then we are left with the following new substrings and starting positions.

$s_1=7$	GTAAACAATATTTATAGC
$s_2=1$	AAAATTTACCTCGCAAGG
$s_3=9$	CCGTACTGTCAAGCGTGG
$s_4=5$	TGAGTAATCGACGTCCCA
$s_5=1$	TACTTCACACCCTGTCAA

Gibbs Sampling: an Example

- 6) We iterate the procedure again with the above starting positions until we cannot improve the score any more.

Gibbs Sampler in Practice

- Gibbs sampling needs to be modified when applied to samples with unequal distributions of nucleotides.
- Needs to be run with many randomly chosen seeds to achieve good results.

Greedy Profile Motif Search vs. Gibbs Sampling

STOP and Think: What are the differences?

Overview of Gibbs Sampling

Unlike **RandomizedMotifSearch**, **Gibbs sampling** will change only a single k -mer in each step, as well as changing this k -mer more liberally.

t t acct t **aac**
 gat **a**t ct gt c
acggcgt t cg !
 ccct **aa**gag
 cgt c**aga**ggt

t **t** acct t aac
 gat **a**t **c**t gt c
 acggcgt **t t** cg
 ccct aa**ag**ag
cgt cagaggt

t t acct t **aac**
 gat **a**t ct gt c
acggcgt t cg !
 ccct **aa**gag
 cgt c**aga**ggt

t t acct t **aac**
 gat at ct **t** **g**t c
acggcgt t cg
 ccct **aa**gag
 cgt c**aga**ggt

RANDOMIZEDMOTIFSEARCH
 (may change all k -mers in one step)

GIBBSAMPLER
 (changes one k -mer in one step)

Gibbs Sampling in Action

Say that we pick the red strings as our *Motifs* of length $k = 4$. Gibbs sampling randomly selects one of the strings to be replaced.

	t t ACCT t aac		t t ACCT t aac
	gAT GTct gt c		gAT GTct gt c
<i>Dna</i>	ccgG CGTt ag	–!	-----
	c act a ACGAg		c act a ACGAg
	cgt cag AGGT		cgt cag AGGT

Gibbs Sampling in Action

Adding pseudocounts allows us to compute a new profile matrix using just the $t - 1$ strings that are remaining.

We then find $\Pr(\text{text} | \text{Profile})$ for every 4-mer in the removed string CCGGCGTTAG.

We now have a new collection of *Motifs* after choosing one based on this “weighted die roll” to replace the one we had removed.

Gibbs Sampling in Action

We now have a new collection of *Motifs* after choosing one based on this “weighted die roll” to replace the one we had removed.

	t t ACCT t aac		ttACCT taac
	gAT GTct gt c		gAT GTct gtc
<i>Dna</i>	ccgGCGT t ag	–!	ccg GCGT tag
	cact a ACGAg		c acta ACGAg
	cgt cag AGGT		cgtcag AGGT

Running these steps N times for some parameter N yields the Gibbs sampler algorithm.

Gibbs Sampling Pseudocode

GibbsSampler(Dna, k, t, N)

randomly select k -mers $Motifs = (Motif_1, \dots, Motif_t)$ from Dna

$BestMotifs \leftarrow Motifs$

for $j \leftarrow 1$ to N

$i \leftarrow$ randomly generated integer between 1 and t

$Profile \leftarrow$ profile formed from all $Motifs$ other than $Motif_i$

$Motif_i \leftarrow Profile$ -randomly generated k -mer in Dna

if $Score(Motifs) < Score(BestMotifs)$

$BestMotifs \leftarrow Motifs$

return $BestMotifs$

References

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